EXISTENCE OF A BIVARIANT CLASS ON A CLOSED SUBSCHEME CHARACTERISED BY VANISHING ON THE COMPLEMENTARY COMPONENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0F9H (sources/stacks-chow/pinned-original-source.tex line 9695).

Allowed standard prerequisites (editorial).

- Writing a cycle class as a sum of pushforwards from a cover by two closed subschemes. Same chapter, Lemma labelled lemma-exact-sequence-closed-chow (source line 2991), invoked at proof lines 9734-9736. Supplies the decomposition alpha' = i'_{1,*}alpha'_1 + i'_{2,*}alpha'_2 and the description of the ambiguity of alpha'_2 as the image of the pushforward from X'_1 cap X'_2, which is exactly the ambiguity the hypotheses remove.
- Chern classes of a perfect object and the capping operation of a bivariant class. Same chapter, Section labelled section-pre-derived (source line 9042), used throughout the proof, and Situation labelled situation-setup (source line 1372) in the statement. Fixes the objects P_p(E_2), c_p(E_2) and the bivariant Chow groups in which the characterised class lives.

Source reference. Original definition 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Same-chapter pointer into the author environment situation at source line 1372; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0ESY: sources/stacks-chow/pinned-original-source.tex, source lines 9042–9042. Same-chapter pointer into the author environment lemma at source line 9042; invoked at proof line 9727 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0F94: sources/stacks-chow/pinned-original-source.tex, source lines 2991–2991. Same-chapter pointer into the author environment lemma

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at source line 2991; invoked at proof line 9734 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Let $i_j : X_j \to X$, $j = 1, 2$ be closed immersions such that $X = X_1 \cup X_2$ set theoretically. Let $E_2 \in D(\mathcal{O}_{X_2})$ be a perfect object. Assume*

(1) *Chern classes of E_2 are defined,*
(2) *the restriction $E_2|_{X_1 \cap X_2}$ is zero, resp. isomorphic to a finite locally free $\mathcal{O}_{X_1 \cap X_2}$-module of rank $< p$ sitting in cohomological degree 0.*

Then there is a canonical bivariant class

$$P'_p(E_2), \text{ resp. } c'_p(E_2) \in A^p(X_2 \to X)$$

characterized by the property

$$P'_p(E_2) \cap i_{2,*}\alpha_2 = P_p(E_2) \cap \alpha_2 \quad \text{and} \quad P'_p(E_2) \cap i_{1,*}\alpha_1 = 0,$$

respectively

$$c'_p(E_2) \cap i_{2,*}\alpha_2 = c_p(E_2) \cap \alpha_2 \quad \text{and} \quad c'_p(E_2) \cap i_{1,*}\alpha_1 = 0$$

for $\alpha_i \in \mathrm{CH}_k(X_i)$ and similarly after any base change $X' \to X$ locally of finite type.

Proof. We are going to use the material of Section 0ESY without further mention.

Assume $E_2|_{X_1 \cap X_2}$ is zero. Consider a morphism of schemes $X' \to X$ which is locally of finite type and denote $i'_j : X'_j \to X'$ the base change of i_j. By Lemma 0F94 we can write any element $\alpha' \in \mathrm{CH}_k(X')$ as $i'_{1,*}\alpha'_1 + i'_{2,*}\alpha'_2$ where $\alpha'_2 \in \mathrm{CH}_k(X'_2)$ is well defined up to an element in the image of pushforward by $X'_1 \cap X'_2 \to X'_2$. Then we can set $P'_p(E_2) \cap \alpha' = P_p(E_2) \cap \alpha'_2 \in \mathrm{CH}_{k-p}(X'_2)$. This is well defined by our assumption that E_2 restricts to zero on $X_1 \cap X_2$.

If $E_2|_{X_1 \cap X_2}$ is isomorphic to a finite locally free $\mathcal{O}_{X_1 \cap X_2}$-module of rank $< p$ sitting in cohomological degree 0, then $c_p(E_2|_{X_1 \cap X_2}) = 0$ by rank considerations and we can argue in exactly the same manner. □